In [ ]:
```
# 1 Load and Explore the Dataset
```

In [3]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
df=pd.read_csv("C:/Users/Administrator/Downloads/Heart Disease UCI.csv")
df
```

Out[3]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	tl
0	69	1	0	160	234	1	2	131	0	0.1	1	1	
1	69	0	0	140	239	0	0	151	0	1.8	0	2	
2	66	0	0	150	226	0	0	114	0	2.6	2	0	
3	65	1	0	138	282	1	2	174	0	1.4	1	1	
4	64	1	0	110	211	0	2	144	1	1.8	1	0	
...	...	...	...	...	...	...	...	...	...	...	...	...	
292	40	1	3	152	223	0	0	181	0	0.0	0	0	
293	39	1	3	118	219	0	0	140	0	1.2	1	0	
294	35	1	3	120	198	0	0	130	1	1.6	1	0	
295	35	0	3	138	183	0	0	182	0	1.4	0	0	
296	35	1	3	126	282	0	2	156	1	0.0	0	0	

297 rows × 14 columns

In [22]:
```
df.head(10)
```

Out[22]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal
0	69	1	0	160	234	1	2	131	0	0.1	1	1	0
1	69	0	0	140	239	0	0	151	0	1.8	0	2	0
2	66	0	0	150	226	0	0	114	0	2.6	2	0	0
3	65	1	0	138	282	1	2	174	0	1.4	1	1	0
4	64	1	0	110	211	0	2	144	1	1.8	1	0	0
5	64	1	0	170	227	0	2	155	0	0.6	1	0	2
6	63	1	0	145	233	1	2	150	0	2.3	2	0	1
7	61	1	0	134	234	0	0	145	0	2.6	1	2	0
8	60	0	0	150	240	0	0	171	0	0.9	0	0	0
9	59	1	0	178	270	0	2	145	0	4.2	2	0	2

In [16]:
```
#checking datattypes
df.dtypes
```

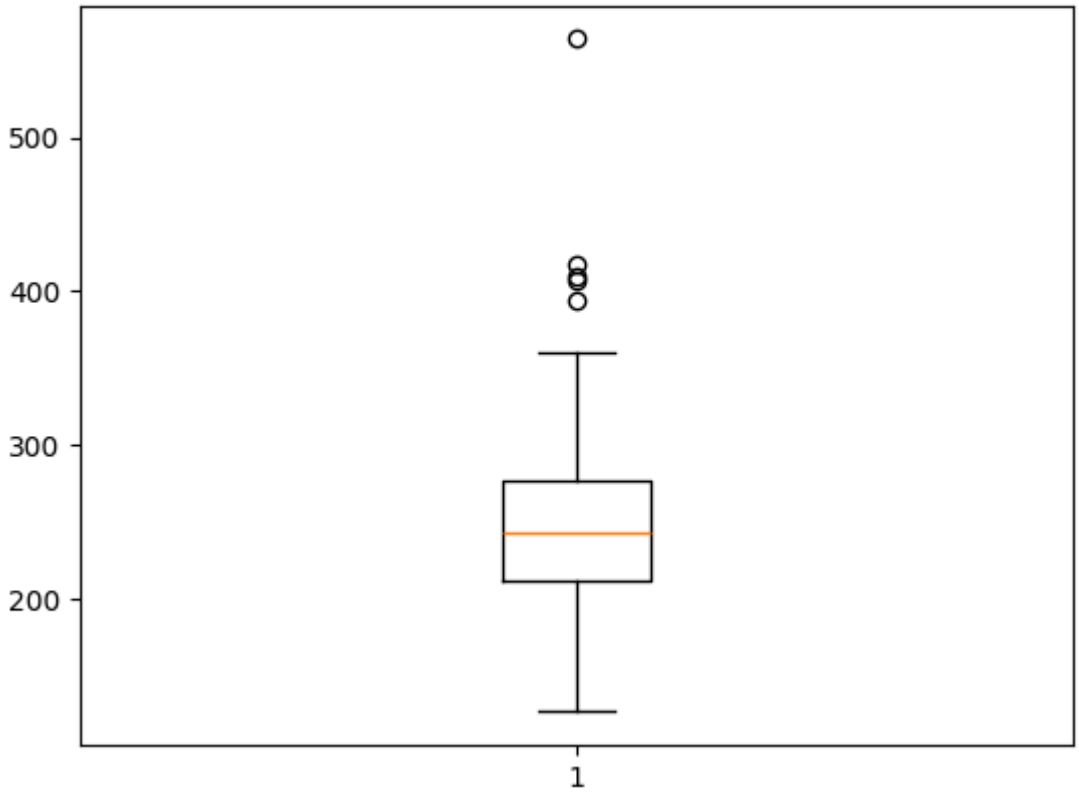
Out[16]:
```
age            int64
sex            int64
cp             int64
trestbps       int64
chol           int64
fbs            int64
restecg        int64
thalach        int64
exang          int64
oldpeak      float64
slope          int64
ca             int64
thal           int64
condition      int64
dtype: object
```

In [13]:
```
#checking if any data is null or finding missing vlues
print("Checking if any data is null or missing")
df.isnull().sum()
```

```
Checking if any data is null or missing
```

Out[13]:
```
age          0
sex          0
cp           0
trestbps     0
chol         0
fbs          0
restecg      0
thalach      0
exang        0
oldpeak      0
slope        0
ca           0
thal         0
condition    0
dtype: int64
```

In [21]:
```
#detecting outliers
import matplotlib.pyplot as plt
plt.boxplot(df['chol'])
plt.show()
```



In [23]:
```
#showing summary statistics
df.describe()
```

Out[23]:

	age	sex	cp	trestbps	chol	fbs	reste
count	297.000000	297.000000	297.000000	297.000000	297.000000	297.000000	297.000
mean	54.542088	0.676768	2.158249	131.693603	247.350168	0.144781	0.996
std	9.049736	0.468500	0.964859	17.762806	51.997583	0.352474	0.994
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.000000	0.000
25%	48.000000	0.000000	2.000000	120.000000	211.000000	0.000000	0.000
50%	56.000000	1.000000	2.000000	130.000000	243.000000	0.000000	1.000
75%	61.000000	1.000000	3.000000	140.000000	276.000000	0.000000	2.000
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.000000	2.000

In [ ]:
```
# 2 Gender Distribution Analysis
```

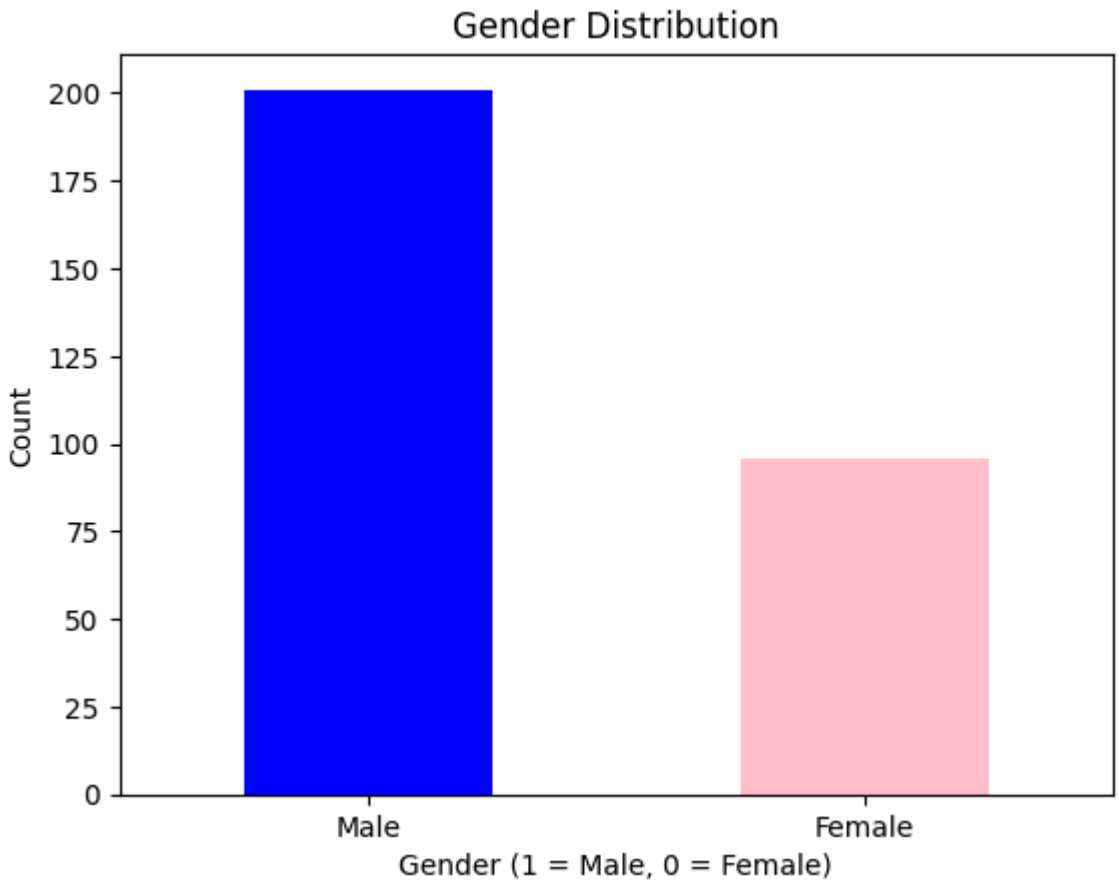
In [101]:
```
total_count = df['sex'].value_counts()
print("Gender Distribution:\n", total_count)

percentage = total_count / total_count.sum() * 100
print("Gender Percentage :\n", percentage)

total_count.plot(kind='bar', color=['blue', 'pink'])
plt.title('Gender Distribution')
plt.xlabel('Gender (1 = Male, 0 = Female)')
plt.ylabel('Count')
```

```
plt.xticks([0, 1], ['Male', 'Female'], rotation=0)
plt.show()
```

```
Gender Distribution:
 sex
1    201
0     96
Name: count, dtype: int64
Gender Percentage :
 sex
1    67.676768
0    32.323232
Name: count, dtype: float64
```



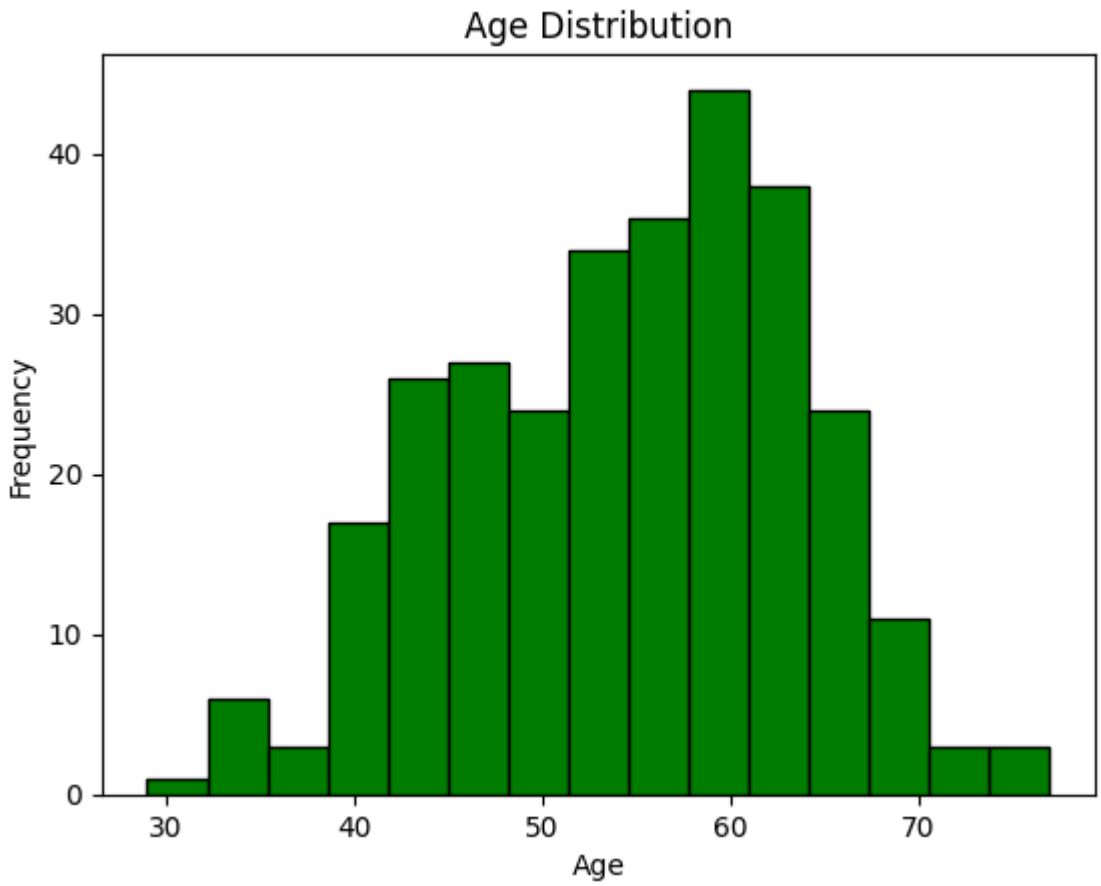
In [ ]:
```
# 3 Age Analysis
```

In [29]:
```
min_age = df['age'].min()
max_age = df['age'].max()
mean_age = df['age'].mean()
median_age = df['age'].median()

print(f"Min Age: {min_age}")
print(f"Max Age: {max_age}")
print(f"Mean Age: {mean_age}")
print(f"Median Age: {median_age}")


d['age'].plot(kind='hist', bins=15, color='green', edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

```
Min Age: 29
Max Age: 77
Mean Age: 54.54208754208754
Median Age: 56.0
```



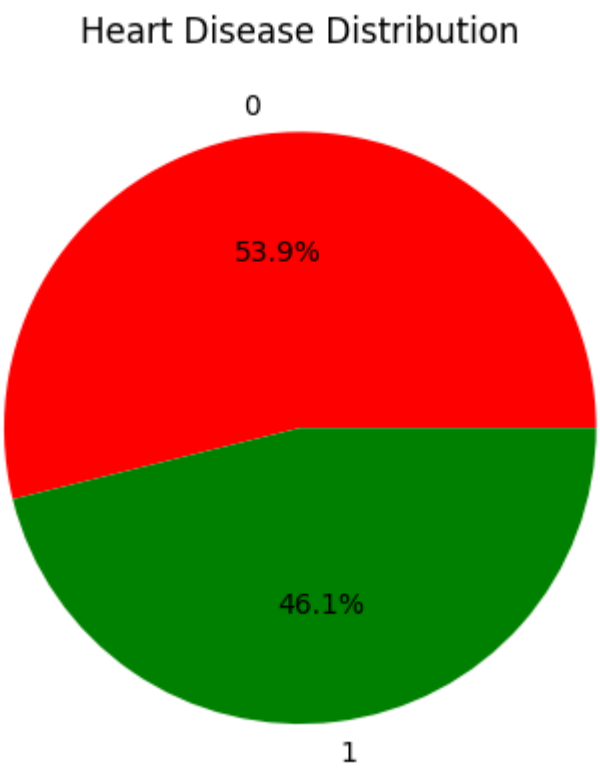
In [ ]:
```
# 4 Target Variable Analysis
```

In [82]:
```
target_counts = df['condition'].value_counts()
print("number of patients with and without heart disease\n", target_counts)

target_counts.plot(kind='pie', autopct='%1.1f%%', colors=['red', 'green'])
plt.title('Heart Disease Distribution')
plt.ylabel('')
plt.show()

disease_percentage = target_counts[1] / target_counts.sum() * 100
print(f"Heart Disease Percentage: {disease_percentage:.2f}%")
```

```
number of patients with and without heart disease
 condition
0    160
1    137
Name: count, dtype: int64
```



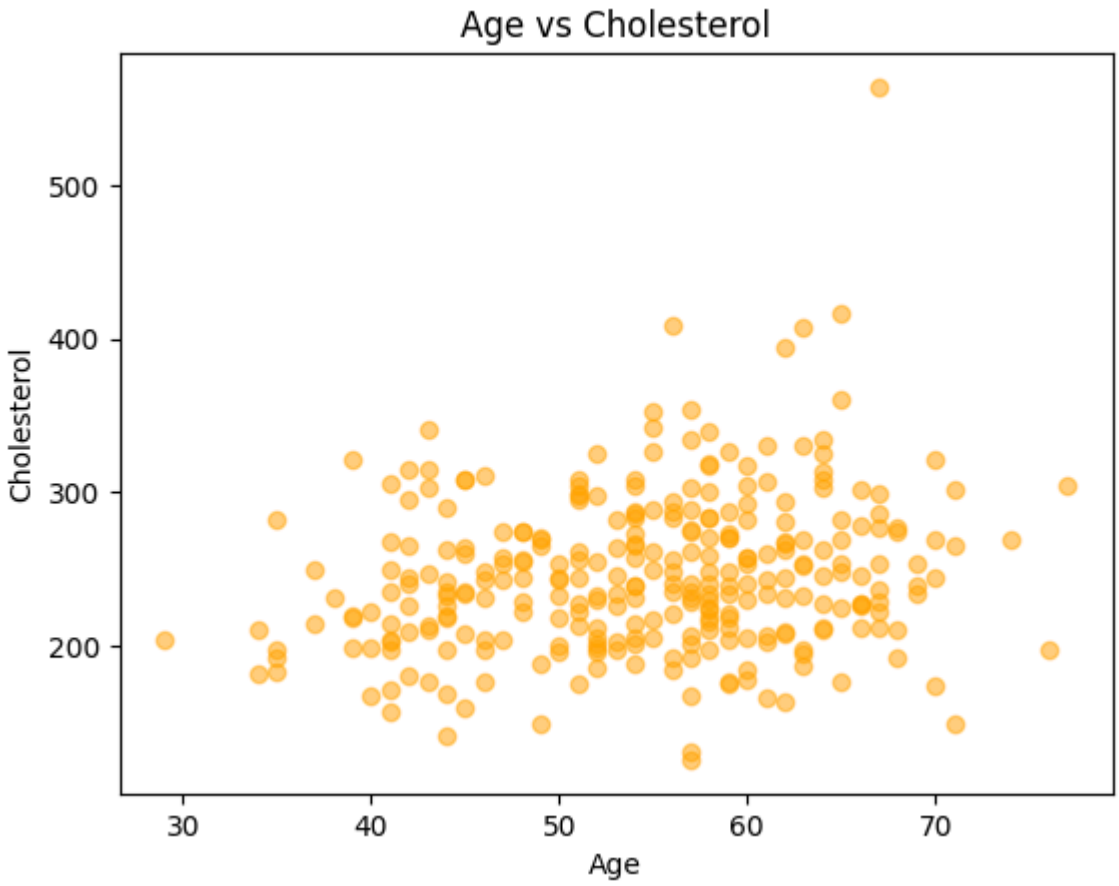

```
Heart Disease Percentage: 46.13%
```

In [ ]:
```
# 5 Correlation Between Age and Cholesterol
```

In [35]:
```
correlation = df[['age', 'chol']].corr().iloc[0, 1]
print(f"Correlation between Age and Cholesterol: {correlation:.2f}")

plt.scatter(df['age'], df['chol'], alpha=0.5, color='orange')
plt.title('Age vs Cholesterol')
plt.xlabel('Age')
plt.ylabel('Cholesterol')
plt.show()
```

```
Correlation between Age and Cholesterol: 0.20
```



In [ ]:
```
#intermediate
```

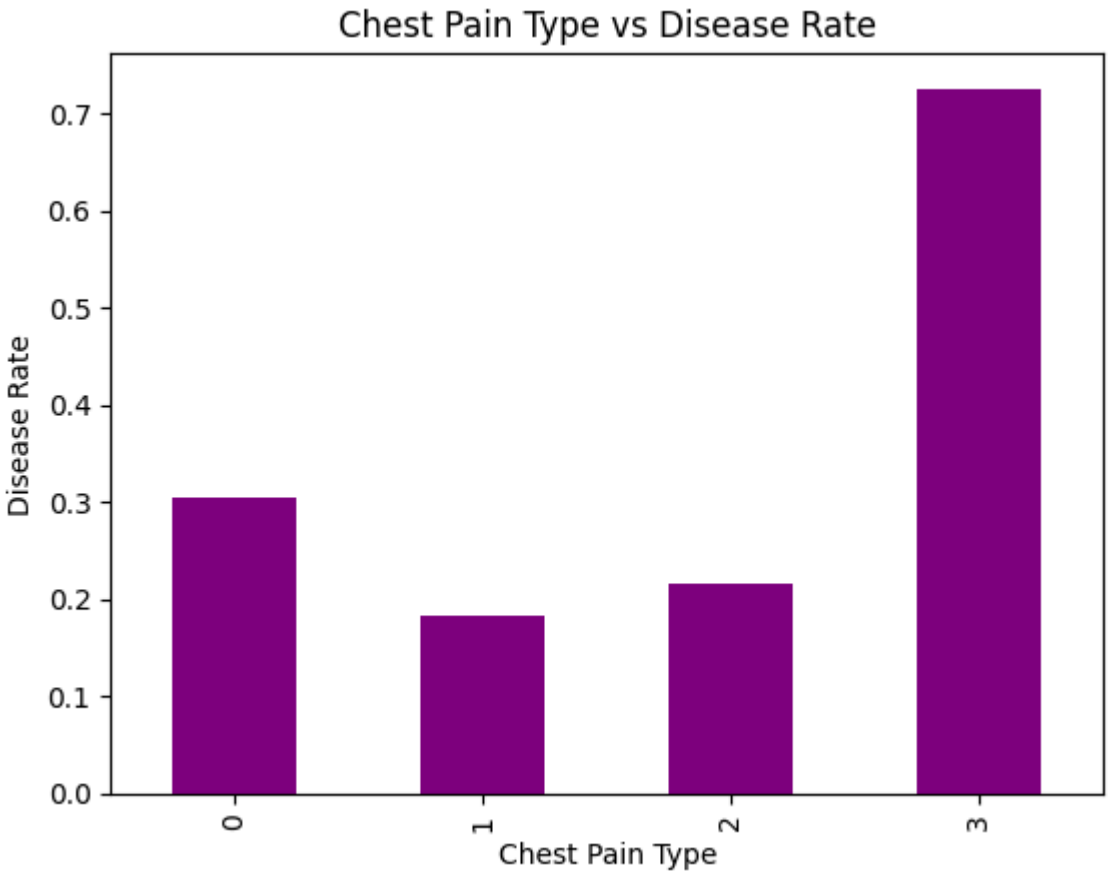
In [ ]:
```
# 6 Chest Pain Type vs Disease
```

In [102]:
```
# multipl by 100
cp= df.groupby('cp')['condition'].mean()
print("Disease Rate by Chest Pain Type:\n", cp)


cp.plot(kind='bar', color='purple')
plt.title('Chest Pain Type vs Disease Rate')
plt.xlabel('Chest Pain Type')
plt.ylabel('Disease Rate')
plt.show()

risky_cp = cp.idxmax()
print(f"The most risky chest pain type is: {risky_cp}")
```

```
Disease Rate by Chest Pain Type:
 cp
0    0.304348
1    0.183673
2    0.216867
3    0.725352
Name: condition, dtype: float64
```



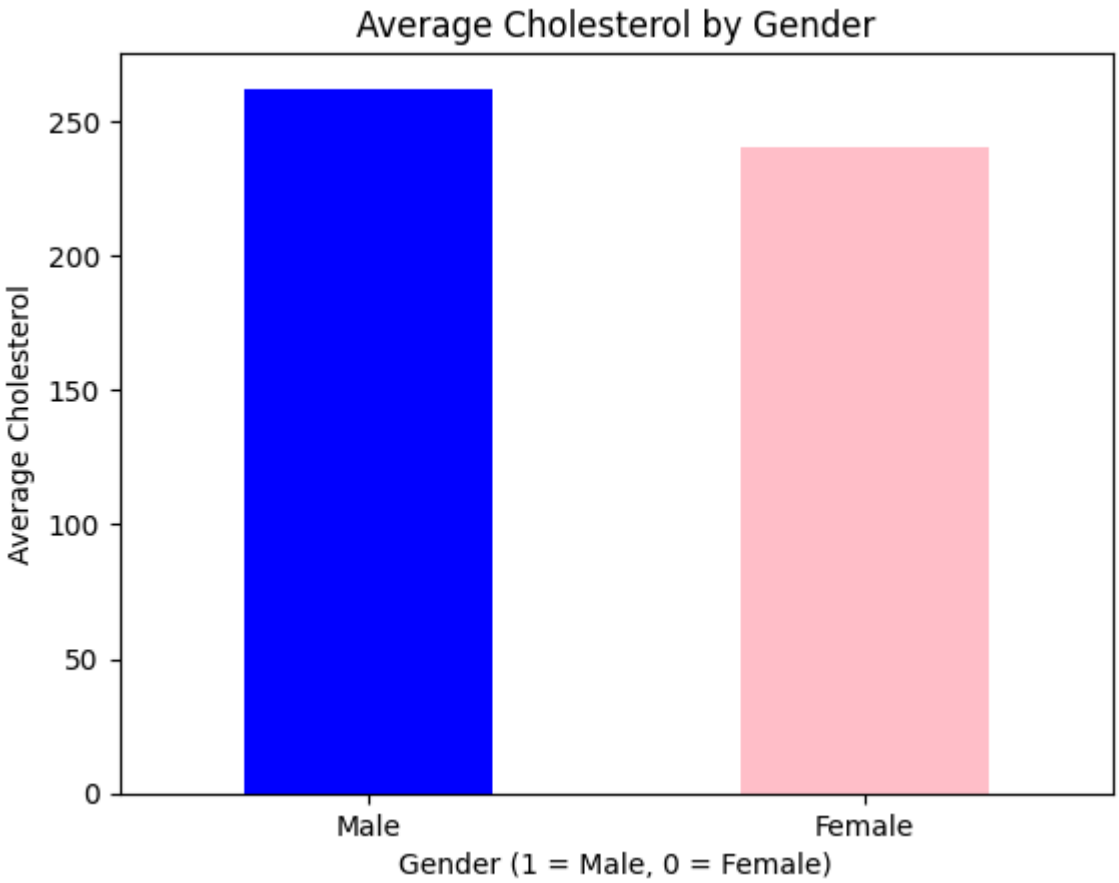

```
The most risky chest pain type is: 3
```

In [ ]:
```
# 7 Average Cholesterol by Gender
```

In [103]:
```
avg= df.groupby('sex')['chol'].mean()
print("Average Cholesterol by Gender:\n", avg)


avg.plot(kind='bar', color=['blue', 'pink'])
plt.title('Average Cholesterol by Gender')
plt.xlabel('Gender (1 = Male, 0 = Female)')
plt.ylabel('Average Cholesterol')
plt.xticks([0, 1], ['Male', 'Female'], rotation=0)
plt.show()
```

```
Average Cholesterol by Gender:
 sex
0    262.229167
1    240.243781
Name: chol, dtype: float64
```



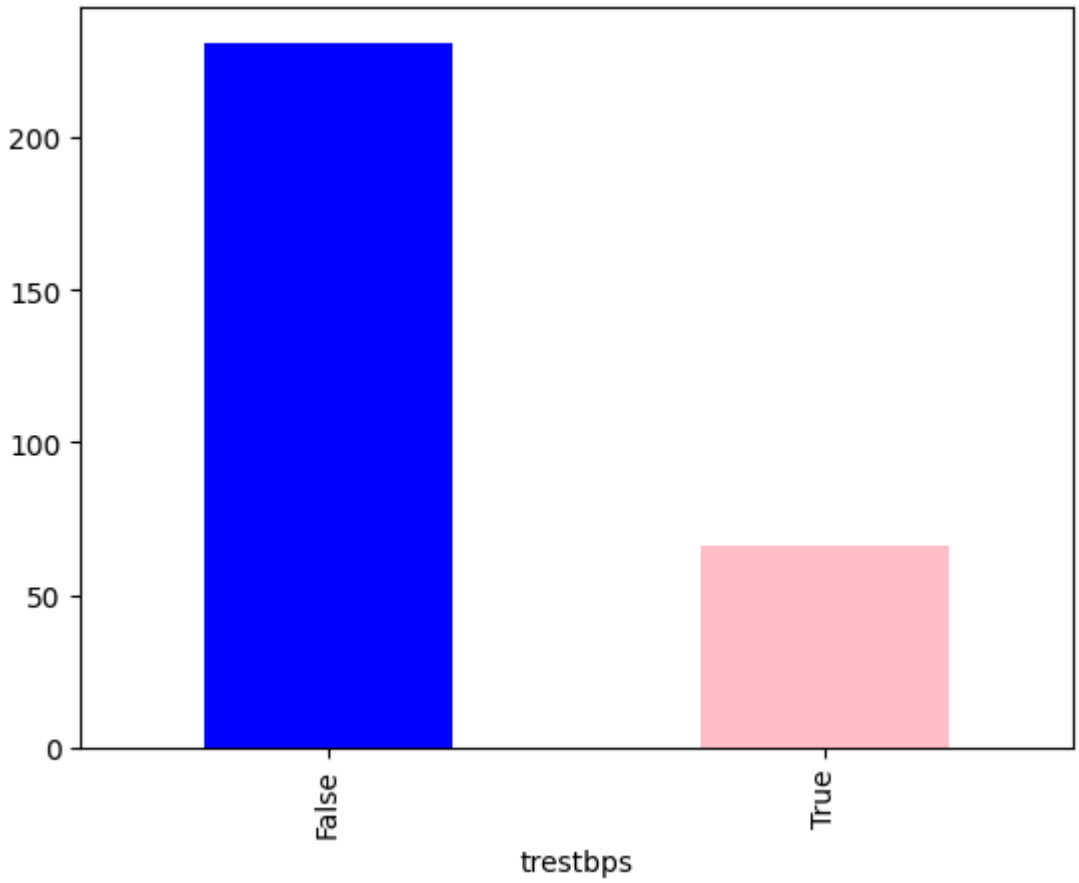
In [ ]:
```
# 8 Resting Blood Pressure Analysis
```

In [110]:
```
avg_bp=np.average((df['trestbps']))
print(f"average bp {avg_bp}")

#patient with bp>140
res=(df['trestbps']>140)
print("patient with bp more than 140 is ",res.sum())
v=res.value_counts()
v.plot(kind='bar', color=['blue', 'pink'])
```

```
average bp 131.69360269360268
patient with bp more than 140 is  66
```

Out[110]:
```
<Axes: xlabel='trestbps'>
```



In [ ]:
```
# 9 Maximum Heart Rate vs Disease
```

In [75]:
```
avg_bp = df['trestbps'].mean()
print("The average bp is",avg_bp)

high_bp = df[df['trestbps'] > 140]
print("the patients with bp more than 140 is :",len(high_bp))


disease= high_bp['condition'].mean()
print(f"disease presence in high BP group: {disease:.2f}")
```

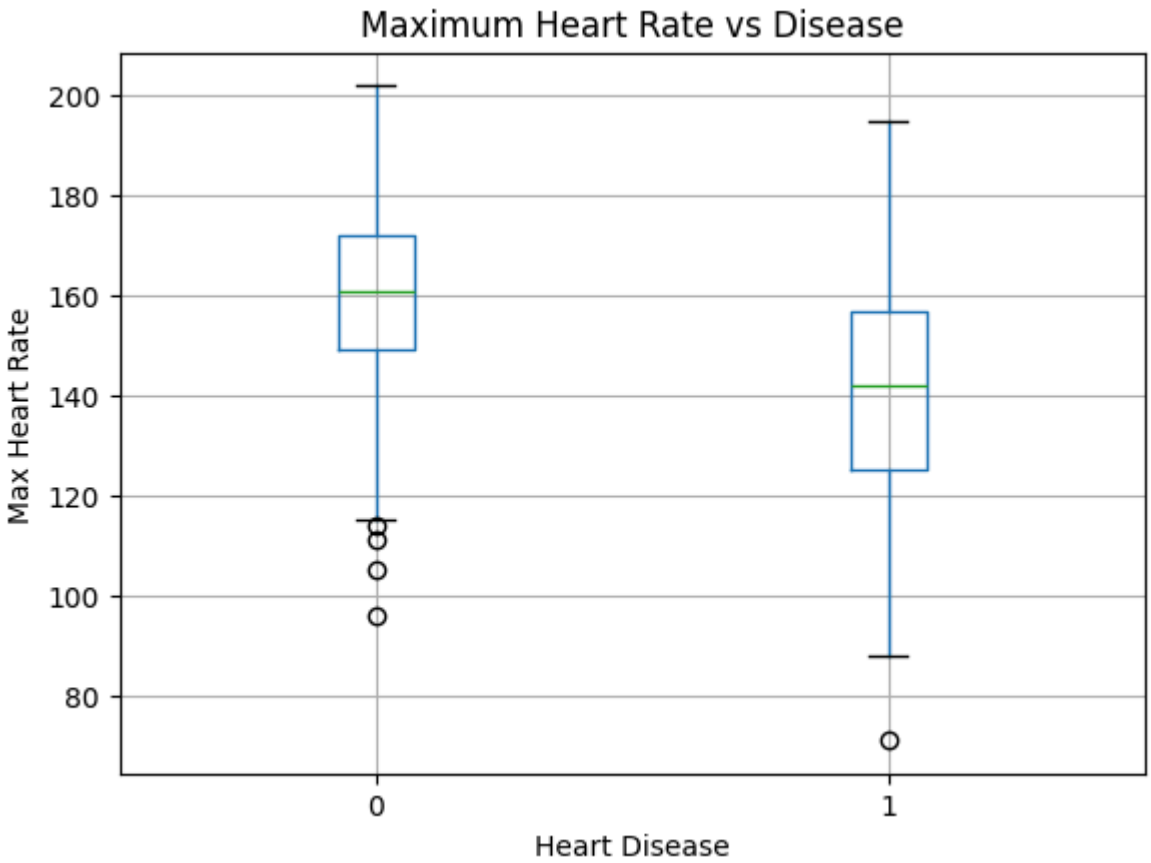
```
The average bp is 131.69360269360268
the patients with bp more than 140 is : 66
disease presence in high BP group: 0.59
```

In [ ]:
```
# 10  Exercise Induced Angina Impact
```

In [77]:
```
disease = df[df['condition'] == 1]['thalach'].mean()
no_disease = df[df['condition'] == 0]['thalach'].mean()

# Plot boxplot
df.boxplot(column='thalach', by='condition')
plt.title("Maximum Heart Rate vs Disease")
plt.suptitle('')
plt.xlabel('Heart Disease')
plt.ylabel('Max Heart Rate')
plt.show()

# Print results
print(f"Average Thalach for Disease: {disease:.2f}")
print(f"Average Thalach for No Disease: {no_disease:.2f}")
```




```
Average Thalach for Disease: 139.11
Average Thalach for No Disease: 158.58
```

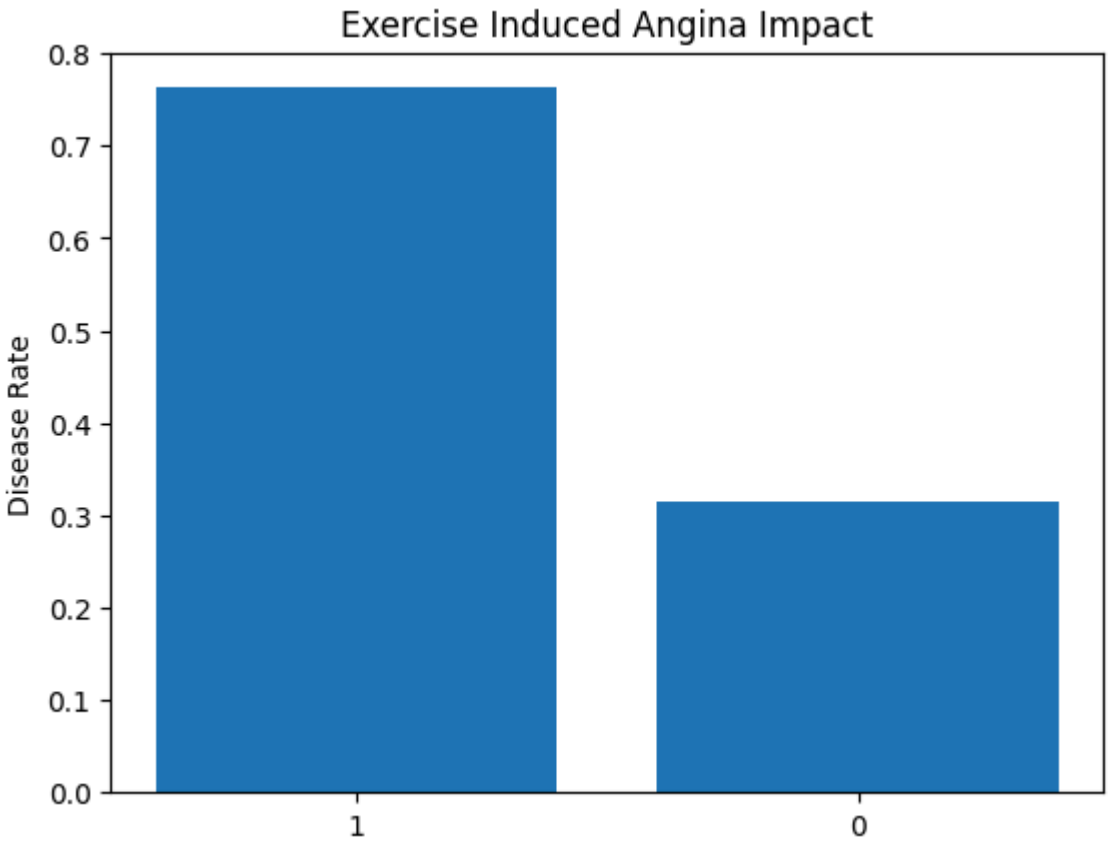
In [ ]:
```
#10. Exercise Induced Angina Impact
```

In [85]:
```
exang_1= df[df['exang'] == 1]['condition'].mean()
exang_0= df[df['exang'] == 0]['condition'].mean()


exang_disease_rates = [exang_1, exang_0]
labels = ['1', '0']

plt.bar(labels, exang_disease_rates)
plt.title("Exercise Induced Angina Impact")
plt.ylabel('Disease Rate')
plt.show()


print(f"Disease rate for exang=1: {exang_1:.2f}")
print(f"Disease rate for exang=0: {exang_0:.2f}")
```



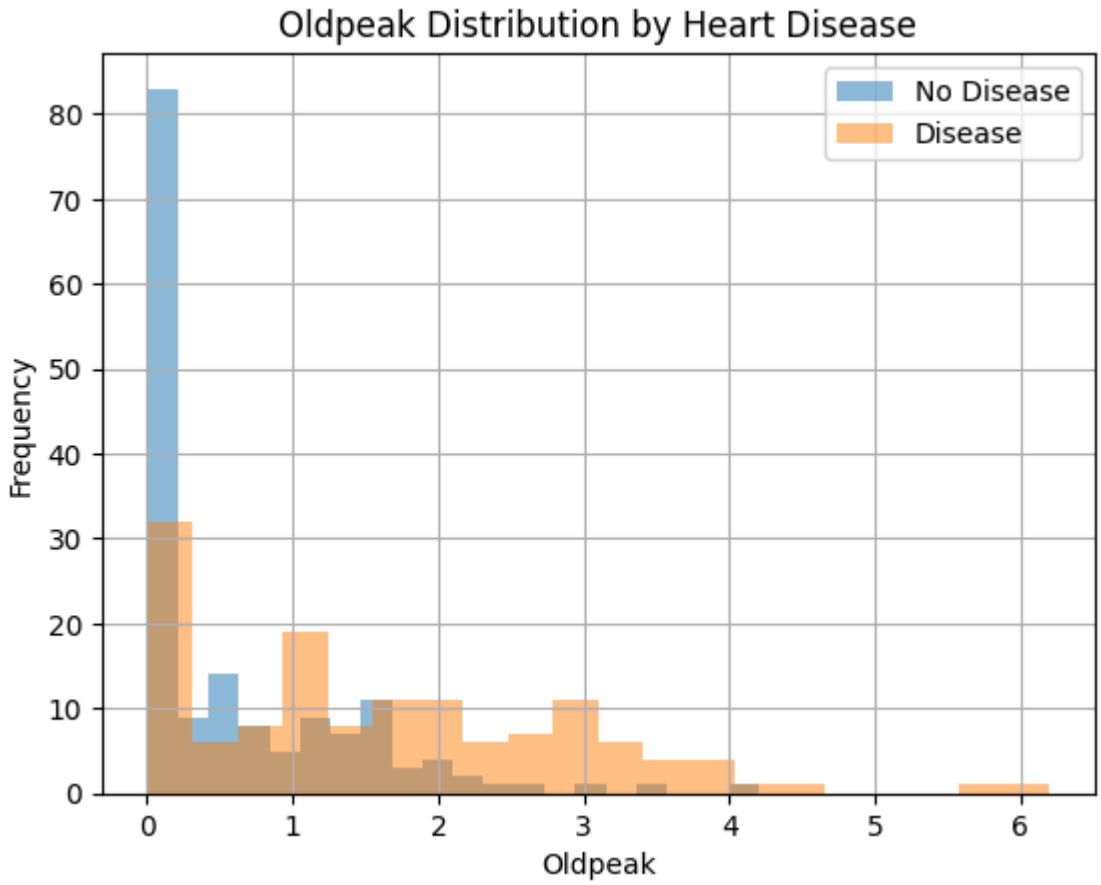

```
Disease rate for exang=1: 0.76
Disease rate for exang=0: 0.32
```

In [ ]:
```
# 11. ST Depression (oldpeak) Analysis
```

In [88]:
```
mean_oldpeak_by_target = df.groupby('condition')['oldpeak'].mean()

df[df['condition'] == 0]['oldpeak'].hist(alpha=0.5, label='No Disease', bins=20)
df[df['condition'] == 1]['oldpeak'].hist(alpha=0.5, label='Disease', bins=20)
plt.legend()
plt.title("Oldpeak Distribution by Heart Disease")
plt.xlabel('Oldpeak')
plt.ylabel('Frequency')
plt.show()


print(mean_oldpeak_by_target)
```

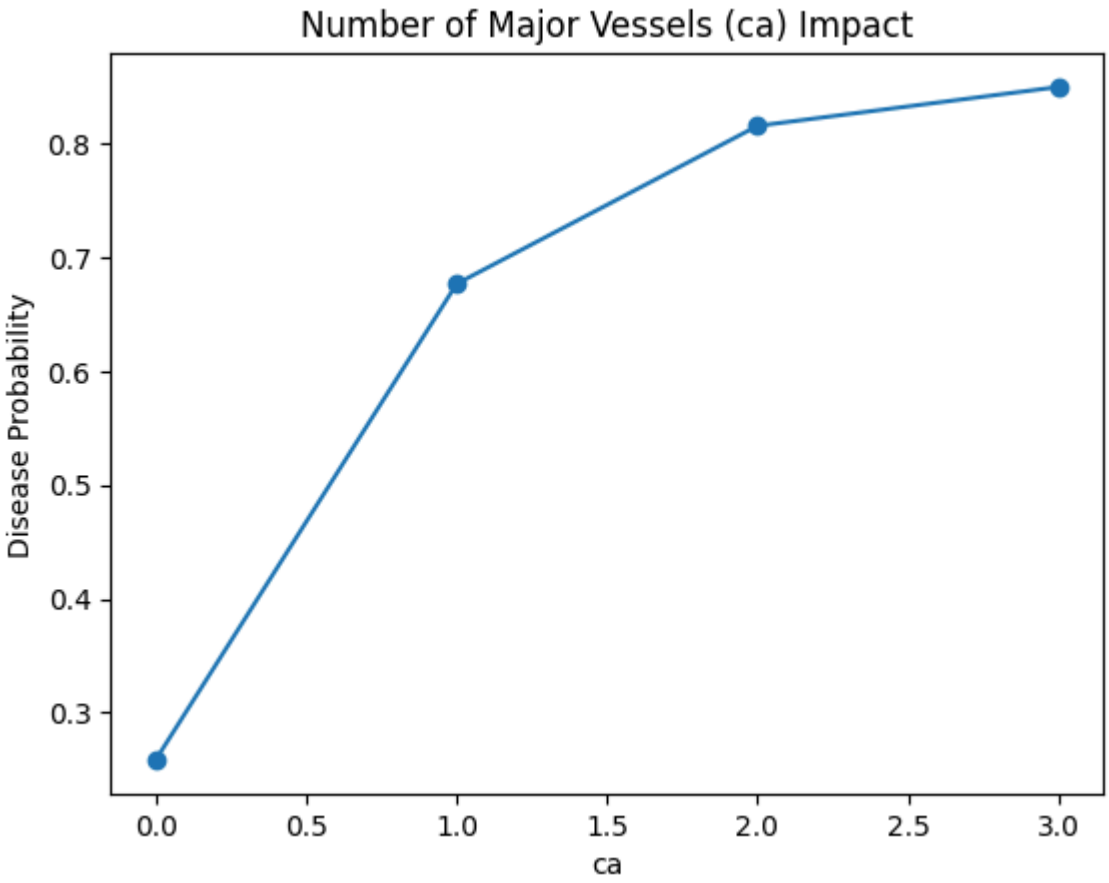



```
condition
0    0.598750
1    1.589051
Name: oldpeak, dtype: float64
```

In [ ]:
```
# Trend of st
```

In [ ]:
```
#12. Number of Major Vessels (ca) Impact
```

In [90]:
```
ca= df.groupby('ca')['condition'].mean()
ca.plot(kind='line', marker='o', title="Number of Major Vessels (ca) Impact")
plt.ylabel('Disease Probability')
plt.show()
```

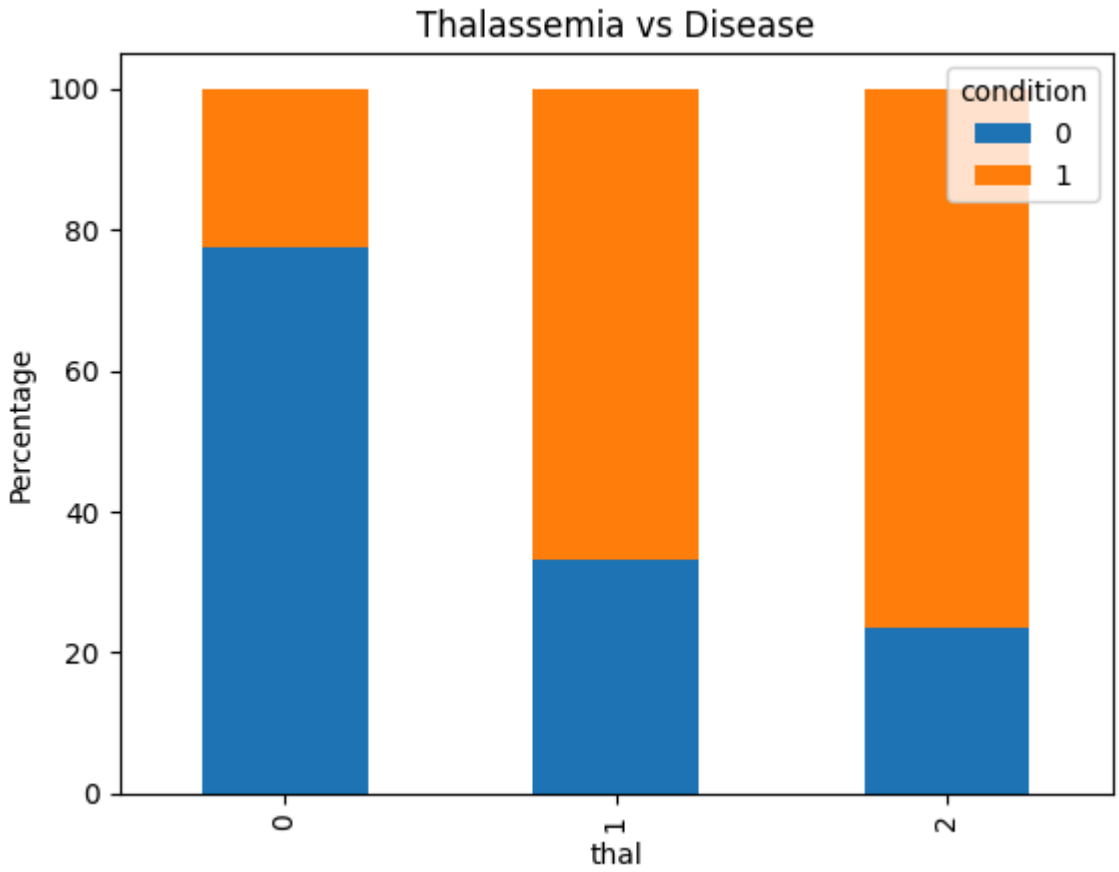


In [ ]:
```
# 13 Thalassemia vs Disease
```

In [93]:
```
cross= pd.crosstab(df['thal'], df['condition'])

cross = cross.div(cross.sum(axis=1), axis=0) * 100

cross.plot(kind='bar', stacked=True, title="Thalassemia vs Disease")
plt.ylabel('Percentage')
plt.show()
```



In [ ]:
```
# 14 Multi-Factor Risk Analysis
```

In [95]:
```
find = df[(df['age'] > 50) & (df['chol'] > 240) & (df['trestbps'] > 140)]
find_percentage = find['condition'].mean() * 100
print(f"High-risk group disease percentage: {find_percentage:.2f}%")
```

```
High-risk group disease percentage: 66.67%
```

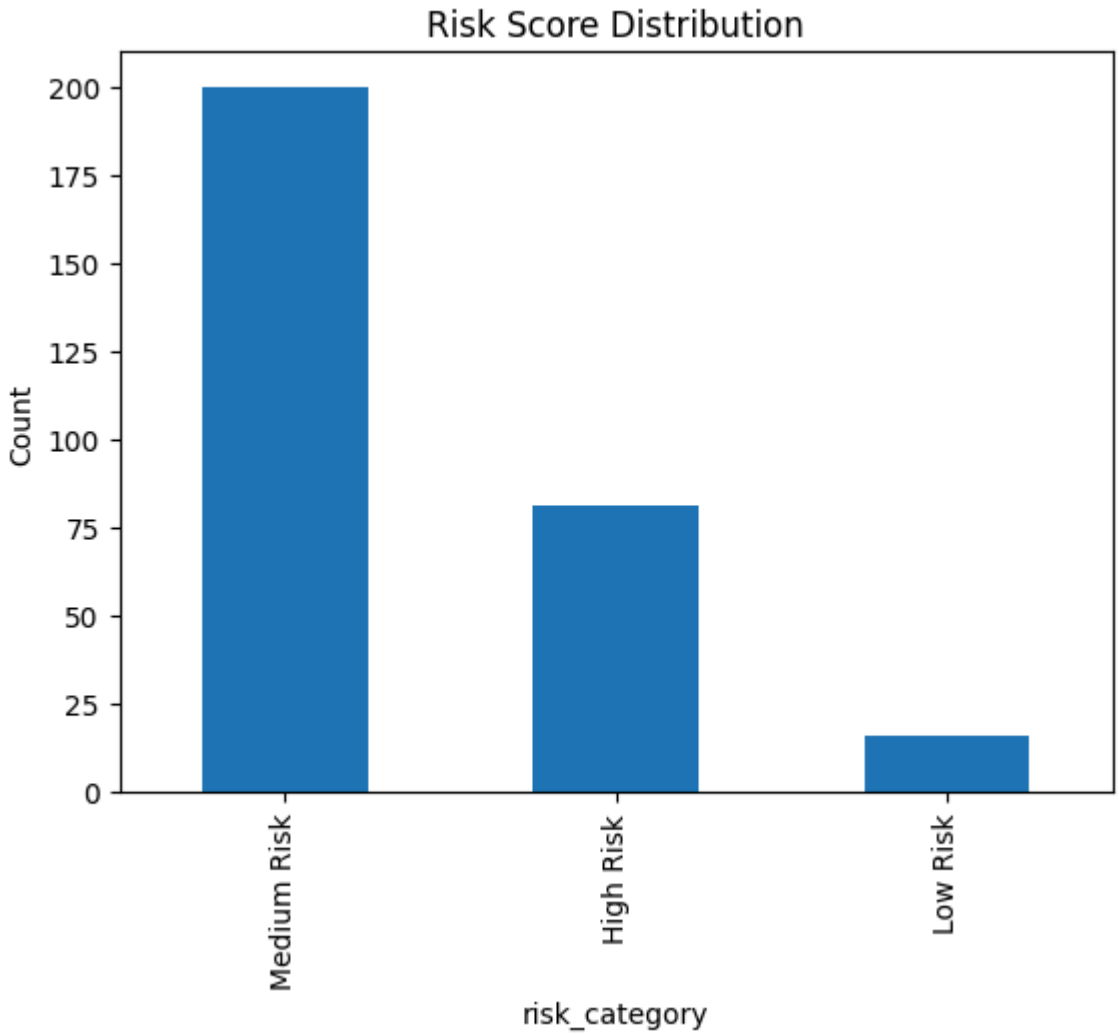
In [ ]:
```
#15. Create Risk Score (Custom Analysis)
```

In [100]:
```
df['risk_score'] = (df['chol'] / 200) + (df['trestbps'] / 120) + df['oldpeak']

conditions = [
    (df['risk_score'] < 2),
    (df['risk_score'] >= 2) & (df['risk_score'] < 4),
    (df['risk_score'] >= 4)
]
choices = ['Low Risk', 'Medium Risk', 'High Risk']

df['risk_category'] = np.select(conditions, choices)


df['risk_category'].value_counts().plot(kind='bar', title="Risk Score Distributior
plt.ylabel('Count')
plt.show()
```



In [ ]:
```
1.Does cholesterol strongly impact heart disease?

No not strongly
Conclusion: Cholesterol contributes to risk but does not strongly impact heart dis

2.Is the male population more vulnerable?
Yes
Male patients show a higher prevalence of heart disease compared to females
Gender-based analysis indicates males are more prone to heart disease
Conclusion:The male population is more vulnerable to heart disease

3.Does exercise-induced angina significantly increase risk?

Yes,
Patients with exercise-induced angina (exang = 1) have a much higher disease perce
This feature shows a clear and strong separation between diseased and non-diseased
Conclusion:Exercise-induced angina is a strong indicator of heart disease risk

4.Which feature has the strongest correlation with disease

Number of major vessels (ca)
ca shows the strongest positive correlation with heart disease
Disease probability increases steadily as the number of major vessels increases
```